

Week 4

4.1 | Measuring performance of AI/Analytics solutions



⌚ 60 minutes

Measuring the performance of AI and analytics solutions is of paramount importance in ensuring their effectiveness and value to businesses. Properly assessing prediction performance not only validates the utility of these solutions but also enables organisations to fine-tune their strategies and optimise decision-making processes. In today's data-driven landscape, where AI-driven insights and recommendations are integral to various industries, accurate performance measurement is the cornerstone of success. It provides organisations with the confidence to rely on these solutions for critical decisions, ultimately driving efficiency, cost-effectiveness, and improved outcomes.

In evaluating the performance of AI and analytics solutions, several simple yet powerful metrics are commonly employed across various machine learning tasks. For classification tasks, metrics like accuracy, precision, recall, and F1-score are indispensable. Accuracy measures the overall correctness of predictions, while precision quantifies the proportion of true positive predictions among all positive predictions. Recall, on the other hand, assesses the fraction of true positives identified correctly from all actual positive instances. The F1-score strikes a balance between precision and recall, providing a single score that summarises the model's performance. For regression tasks, metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared (R²) are fundamental. MAE quantifies the average magnitude of prediction errors, while MSE emphasises larger errors. R², known as the coefficient of determination, gauges the proportion of variance in the target variable that the model explains. These metrics, though simple, serve as essential tools for organisations to gauge the quality and reliability of their AI and analytics solutions, facilitating informed decision-making and continuous improvement.

The challenge

You will build skills in critically evaluating AI and analytics solutions by comparing machine learning model performances. This is crucial for Assessment 1, where you'll be tasked with evaluating the effectiveness of machine learning models in a

business context, providing recommendations for its optimisation and deployment based on performance metrics.

The output



Comparisons of ML models.

The task

This challenge will present to you two business scenarios which show case the applications of two popular machine learning techniques: Linear Regression and Decision Tree. In each scenario, you will see the steps to gain further understanding of the datasets (e.g. via visualisations) and how the ML model (after training) can be interpreted.

In Week 3, we'll dive deeper into these methods and introduce you to a no-code tool designed to simplify these processes and provide you with more details about the training configuration. The scenarios here will introduce you to the process that you will experience next week to build a functional model.

In this challenge, you will apply common metrics to measure the performance of ML models. You will then need to compare the performance of different ML models and recommend the one with the best prediction quality. Two scenarios will be presented in this challenge: (1) House Price Prediction, and (2) Employee Attrition Prediction. You will also see two popular ML techniques in action:

- Linear regression to predict house price
- Decision tree to predict employee attrition

You will apply the equations provided to calculate all metrics MSE, MAE, MAPE, and R2 for House Price Prediction and Accuracy, Precision, Recall, and F1 for Employee Attrition Prediction.

You will then recommend the best model for each scenario. You will need to conduct quick research about the applications of these metrics. Some hints:

- For regression, models with lower MSE, MAE, and MAPE and higher R2 are preferred.
- For classification, models with higher accuracy, prediction, recall, and F1-score are preferred.

Instructions

Review the task materials, then work through the following steps:

1. Download the performance metrics file:



2. Explore the excel file and review the example data for each model and each scenario. For each scenario, you will see the predictions made by each ML model (for example, Linear Regression and Decision Tree Regression for House Price Prediction). Noted that the input data (e.g., location, size, bedrooms) is the same for each ML model, the difference between the predicted values and actual values will decide the prediction quality of each ML model.
3. Select each scenario title to expand the description and complete each accompanying task. Remember to save a copy of your calculated metrics and recommendations for each scenario.

▼ Scenario 1: House price prediction

In this first scenario, the goal is to predict the house price based on some common features:

- Size (in square feet): The total area of the house.
- Number of Bedrooms: Total bedrooms in the house.
- Age of the House (in years): How old the house is.
- Location Type: Categorise the location as 'Urban', 'Suburban', or 'Rural'.
- Price (in USD): The price of the house.

The following infographic depicts visualisations of the dataset:

- Scatter plots for continuous variables (Size, Age) against Price.

- Box plots to see how Price varies with categorical variables (Number of Bedrooms, Location).

Select the plus (+) icons in the interactive to expand insights about each graph.



A data scientist is tasked to develop a linear regression model to predict the house price:

$$\text{Price} = \beta_0 + \beta_1 \times \text{Size} + \beta_2 \times \text{Bedrooms} + \beta_3 \times \text{Age} + \beta_4 \times \text{Location}_1 + \beta_5 \times \text{Location}_2 + \dots + \epsilon$$

Here:

- β_0 is the y-intercept (the model's prediction when. all features are 0).
- $\beta_1, \beta_2, \dots$ are the coefficients for each feature, representing the impact of each feature on the house price.

- $Location_1, Location_2$... are the one-hot encoded variables for the location.
- ϵ is the error term.

Running the linear regression algorithm, the data scientist found the below model:

$$\begin{aligned} \text{Price(USD)} = & 48572.77 + 3803.02 \times \text{Location_Rural} - \\ & 13299.58 \times \text{Location_Suburban} + \\ & 9496.56 \times \text{Location_Urban} + 152.92 \times \text{Size(sqft)} - \\ & 4348.76 \times \text{Bedrooms} - 595.79 \times \text{Age(years)} \end{aligned}$$

Some insights from the model:

- **Size Matters – A Lot:** For every additional square foot, prices increase by **\$152.92**. Bigger houses command higher prices, making size the most reliable factor in predicting value. If you're building or flipping properties, maximising usable space is your golden ticket!
- **Location, Location, Location – But Not the Whole Story:** **Urban homes** gain an extra **\$9,496** compared to the baseline, while **rural homes** enjoy a modest **\$3,803** boost. However, **suburban properties** come with a surprising **\$13,299 drop**—possibly reflecting buyers' changing preferences toward urban or rural living. Location impacts price, but it's not the whole puzzle.
- **Bedrooms do not Contribute much to the Price:** Adding extra bedrooms lowers the price by **\$4,348 per bedroom**. This not a big number and might be the problems with the diverse houses considered in this dataset. Additional features such as on suite, style, etc. may help to explain the price better. More work on data collection will be needed.
- **The Age Factor – A Slow Decline:** Older homes lose **\$595 per year**, reflecting depreciation and increased maintenance. But this is a gradual decline, signalling that age isn't a dealbreaker.

Visualise results:



This plot compares the predicted house prices from the linear regression model against the actual prices from the test set. Points that lie along the dashed line represent accurate predictions where the predicted price matches the actual price.

Task

Now, we will apply the data from the performance metrics spreadsheet.

1. See the subset of the data with predicted values in the performance metrics file and calculate the following metrics:

1. Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

2. Mean Absolute Error

$$MSE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

3. Mean Absolute Percentage Error (MAPE)

$$MSE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100\%$$

4. R-squared (R^2)

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

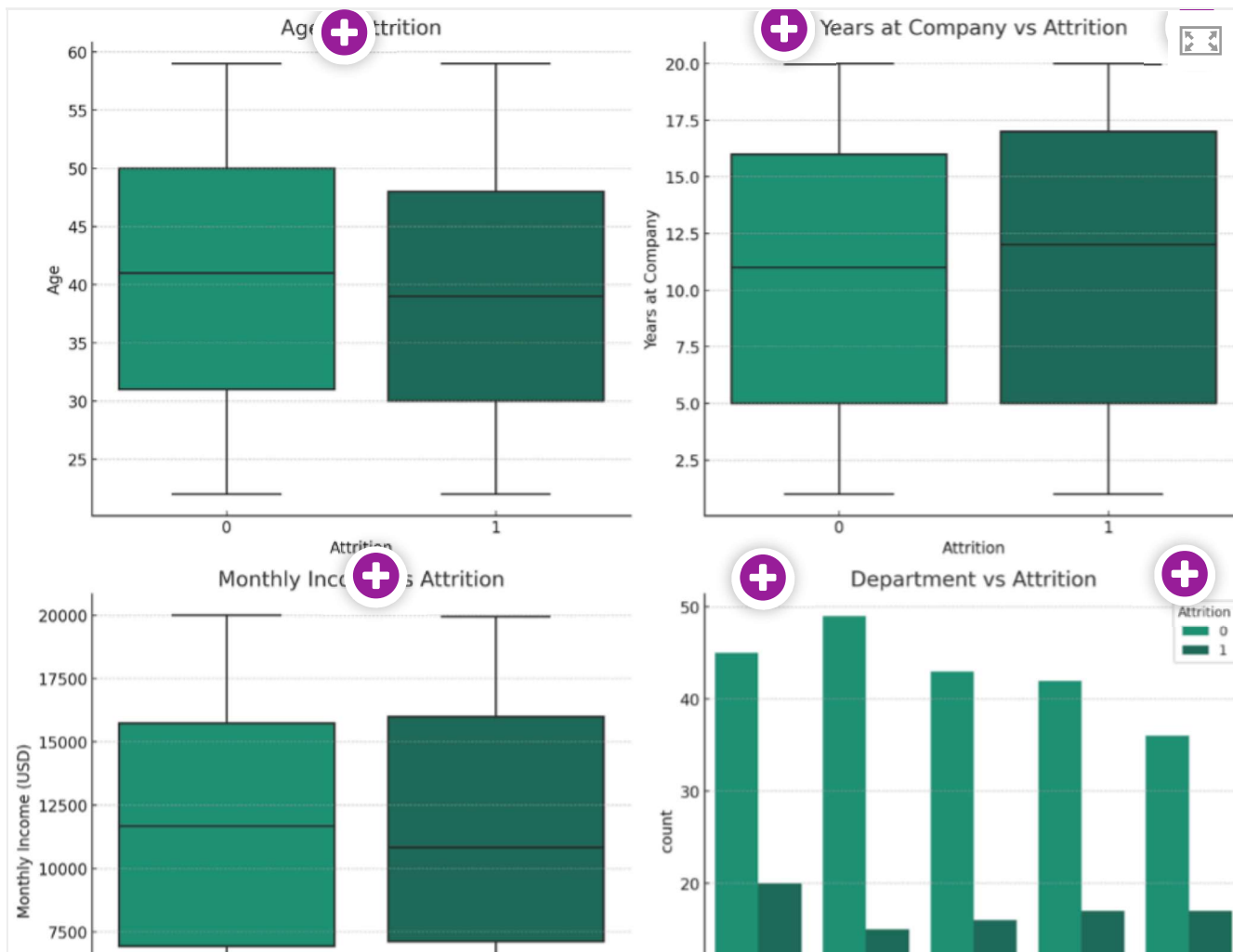
2. Perform the same calculation for the decision regression tree model and suggest which model we should use.

▼ Scenario 2: Predict employee attrition

In this second scenario, we'll need to consider various factors that might influence an employee's decision to leave a company. Here are some potential features for the dataset:

- Age: The age of the employee.
- Department: The department in which the employee works (e.g., Sales, HR, Tech).
- Job Satisfaction: A rating of the employee's job satisfaction (e.g., Low, Medium, High).
- Years at Company: How many years the employee has been with the company.
- Distance from Home: The distance from the employee's home to the workplace.
- Monthly Income: The employee's monthly income.
- Attrition: Whether the employee has left the company (Yes or No).

Review visualisations of the features and attrition used in this scenario by exploring the following interactive. Select the plus (+) icons to expand insights about each graph.



A data scientist is asked to build a decision tree model to predict employee attrition. A part of this model is presented below:

Rule 1: IF Job Satisfaction is Low (≤ 0.50) AND Distance from Home (km) ≤ 49.19 AND Monthly Income (USD) ≤ 3081.56 THEN Prediction: Left

Rule 2: IF Job Satisfaction is Low (≤ 0.50) AND Distance from Home (km) ≤ 49.19 AND Monthly Income (USD) > 3081.56 THEN Prediction: Stayed

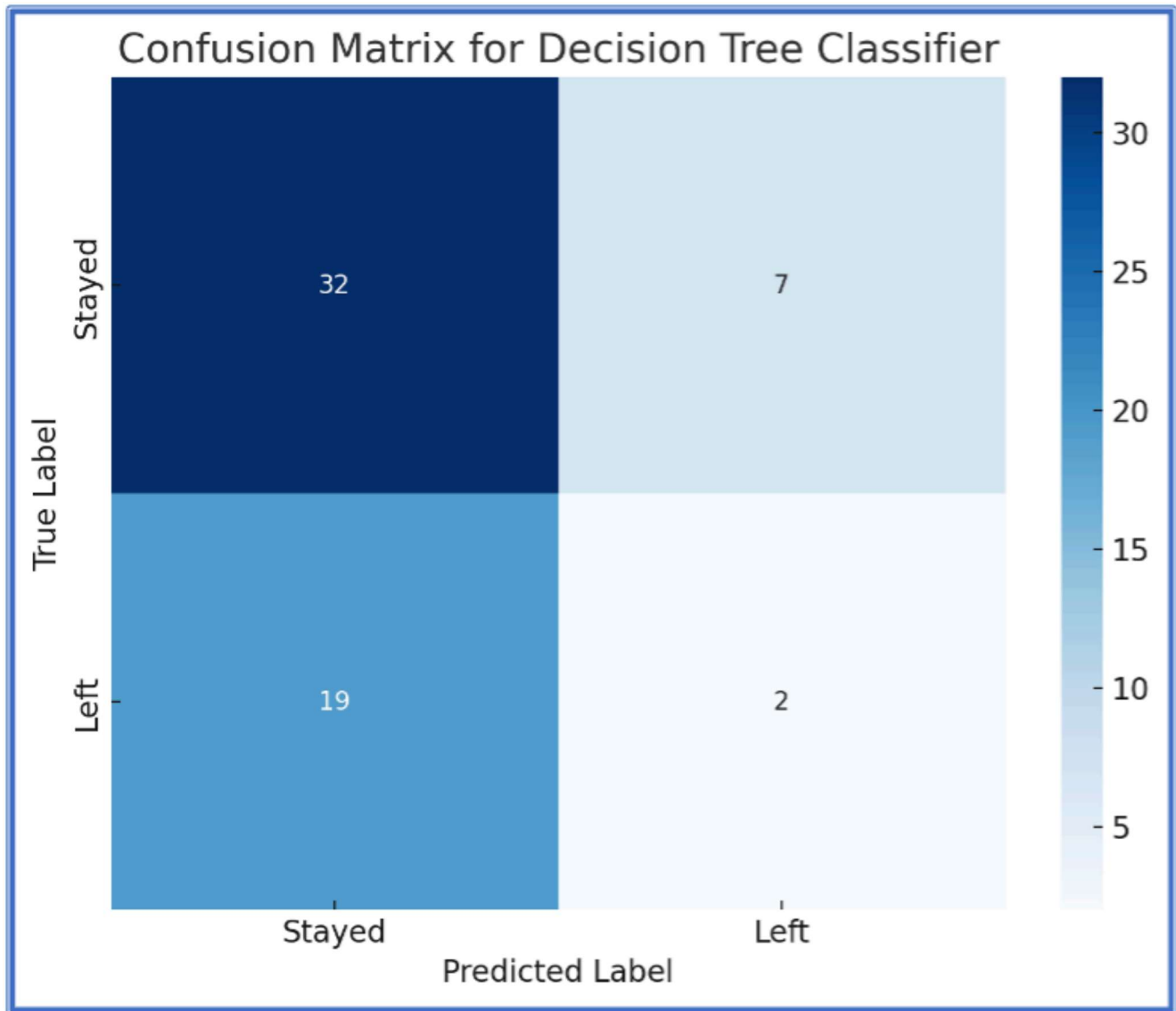
Rule 3: IF Job Satisfaction is Low (≤ 0.50) AND Distance from Home (km) > 49.19 AND Distance from Home (km) ≤ 49.90 THEN Prediction: Left

...

Interpretation: Employees with **low job satisfaction** are more likely to leave unless certain conditions encourage them to stay. If they live **close to the office (within 49.19 km)** and earn a **higher income (above \$3,081.56/month)**, they tend to stay despite their dissatisfaction. However, those with **lower income** or who live **slightly farther away**

(between 49.19 and 49.90 km) are more likely to leave. Overall, **income and commuting distance** can moderate the impact of low job satisfaction on attrition.

Below is the visualisation of a confusion matrix showing if the model makes good predictions:



Task:

Now, we will apply the data from the performance metrics spreadsheet.

1. See the subset of the data with predicted values in the performance metrics Excel spreadsheet and calculate the following metrics:

Accuracy:

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{True Positives} + \text{True Negatives} + \text{False Positives} + \text{False Negatives}}$$

Precision:
$$\text{Precision} = \text{True Positives} / (\text{True Positives} + \text{False Positives})$$
Recall (Sensitivity or True Positive Rate):
$$\text{Recall} = \text{True Positives} / (\text{True Positives} + \text{False Negatives})$$
F1-Score:
$$\text{F1-Score} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$$

In these formulas:

- True Positives (TP) are the number of correctly predicted positive instances.
- True Negatives (TN) are the number of correctly predicted negative instances.
- False Positives (FP) are the number of incorrectly predicted positive instances.
- False Negatives (FN) are the number of incorrectly predicted negative instances.

2. An expert also proposed a rule to predict attribution. Is the expert rule better or worse than the decision tree?



Be prepared to share your work and justify your choice in the live class.

Watch: Performance Metric Recap

Regression Scenario: House price prediction (2)

Linear Regression model

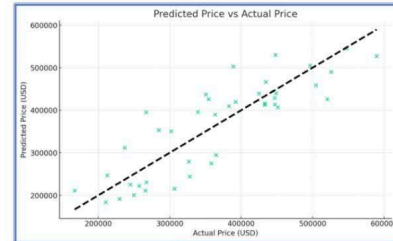
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Supporting resources

- In [2.2 | Mapping machine learning tasks](#), you watched two videos from the LinkedIn Learning module Azure AI Fundamentals (AI-900) Cert Prep: 2 Principles of Machine Learning on Azure. Completing this module will help you to strengthen your ML understanding and Azure ML. Access the module through your Leganto reading list.